Confidence Intervals & Hypothesis Testing

Section 1: Constructing & Interpreting Confidence Intervals

1 What is the correct interpretation of a 95% confidence interval for a population mean $(42.1, 48.7)$?

a. We are 95% confident that the true population mean falls between 42.1 and 48.7

b. There is a 95% probability that the true mean is between 42.1 and 48.7

c. 95% of all individual values in the population lie between 42.1 and 48.7

d. 95% of sample means will equal the population mean

2 In a sample of 100 items, $\hat{p} = 0.60$. With critical value $z^* = 1.96$, what is the margin of error for a 95% confidence interval for p?

a. $1.96\sqrt{\frac{0.60(0.40)}{100}} \approx 0.096$

b. $1.96\sqrt{\frac{0.50(0.50)}{100}} \approx 0.098$

c. 0.60 ± 0.05

d. 0.049

3 How can you decrease the width of a confidence interval?

a. Increase the sample size or decrease the confidence level

b. Increase the confidence level or increase the sample size

c. Decrease the sample size or increase the confidence level

d. Increase the standard deviation of the data

Section 2: Hypothesis Testing Logic & P-values

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4 What is the statistical definition of a P-value?

a. The probability of observing a test statistic as extreme or more extreme than the one observed, assuming the null hypothesis is true

b. The probability that the null hypothesis is true

c. The probability that the alternative hypothesis is false

d. The probability of making a Type I error

5 If a test yields a P-value of 0.023 and the significance level is $\alpha = 0.05$, what is the conclusion?

a. Reject H_0; there is convincing statistical evidence in favor of H_a

b. Fail to reject H_0; the evidence is insufficient

c. Accept H_0 as definitively proven

d. Reject H_a; the test is inconclusive

6 Why do statisticians state 'fail to reject H_0' instead of 'accept H_0'?

a. Failing to find sufficient evidence against H_0 does not prove that H_0 is true

b. Because H_0 is always false in practice

c. To avoid computing the test statistic

d. Because the P-value is always approximate

Section 3: Type I Error, Type II Error, and Power

7 What is a Type I error?

a. Rejecting the null hypothesis when the null hypothesis is actually true

b. Failing to reject the null hypothesis when the null hypothesis is actually false

c. Calculating an incorrect test statistic

d. Using the wrong degrees of freedom

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8 What is a Type II error?

a. Failing to reject the null hypothesis when the null hypothesis is actually false

b. Rejecting the null hypothesis when it is true

c. Selecting a biased sample

d. Misinterpreting the P-value

9 What is the power of a statistical test?

a. The probability of correctly rejecting a false null hypothesis ($1 - \beta$)

b. The probability of making a Type I error (α)

c. The sample size n

d. The probability of retaining a true null hypothesis

Section 4: One-Sample and Two-Sample t-Procedures

10 When estimating a population mean μ with unknown σ, which distribution and test statistic are used?

a. t-distribution with $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and $df = n - 1$

b. Standard normal distribution with $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

c. Chi-square distribution with $df = n - 1$

d. F-distribution with $df = (n - 1, n - 2)$

11 How does the t-distribution compare to the standard normal (Z) distribution?

a. The t-distribution is symmetric and bell-shaped but has heavier tails (more spread)

b. The t-distribution is skewed right

c. The t-distribution has less variability than Z

d. The t-distribution is bounded between -3 and 3

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12 A researcher compares scores before and after a training program on the same 20 subjects. Which test should be performed?

a. Paired t-test on the differences ($d_i = \text{After}_i - \text{Before}_i$)

b. Two-sample independent t-test

c. Two-sample z-test for proportions

d. Chi-square test of independence

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